

CSE440: Natural Language Processing II

Lab Assignment 3

Problem 1

- Load the Iris dataset from `sklearn.datasets.load_digits()`.
- Split the data into 75 % training and 25 % test sets.
- Define a feed-forward neural network with:
 - 2 hidden layers
 - First hidden layer: 64 units, ReLU activation
 - Second hidden layer: 32 units, ReLU activation
 - Output layer: Number of units equal to the number of classes, softmax activation
 - Optimizer: Adam
 - Loss: Categorical cross-entropy
 - Epochs: 150
- Train the network on the training set.
- Print the model summary and the accuracy on the test set.
- Plot the training and validation loss (or accuracy) curves across epochs.

Problem 2

- Use `keras.datasets.imdb` to load the IMDB movie-review dataset. Limit the vocabulary to the top 8,000 most frequent words.
- Take the original training set and split it further into 80 % training and 20 % validation.
- Define a neural network with:
 - Use an embedding layer (8,000 input dim, 50 output dim), then a dense layer with 24 ReLU units, and a sigmoid output layer with 1 unit.

- Loss: Binary cross-entropy.
- Optimizer: SGD
- Epochs: 15
- Train the model on the training set.
- Evaluate the model on the official **test data** provided in `keras.datasets.imdb`.
 - Print the summary of your network.
 - Report the test accuracy.
 - Plot the training and validation loss and accuracy curves.
 - Draw the ROC curve and calculate the AUC (area under the curve) to evaluate the model's performance beyond just accuracy.

Problem 3

- Reuse the preprocessed data from Question 2.
- Take the original training set and split it into 80 % training and 20 % validation.
- Use an embedding layer (output dimension = 64) followed by three hidden dense layers:
 - Hidden Layer 1: 48 units, ReLU, Dropout = 0.3.
 - Hidden Layer 2: 24 units, ReLU, Dropout = 0.2.
 - Hidden Layer 3: 12 units, ReLU.
 - Output layer: 1 unit, sigmoid activation.
 - Loss: Binary cross-entropy.
 - Optimizer: SGD
 - Epochs: 12-15.
- Train the model on the training set, then compare the shallow model from Question 2 with the deep model in terms of training time, test accuracy, overfitting (difference between training and validation performance), and confusion matrices. Summarize which model performs better overall and justify your conclusion.

Problem 4

Reuse the entire setup from problem 3, use learning rate = 0.01 and 0.0001 to compare the model performance.